

Differentiating Preschoolers' Sequential Planning Ability From Their General Intelligence: A Study of Organization, Systematic Responding, and Efficiency in Young Children

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Preschoolers' sequential planning ability was measured in three school-related tasks by assessing organization, systematic responding, and efficiency. Intelligence was assessed through the K-ABC, PPVT, and a Piagetian task. A factor analysis showed evidence of a sequential planning factor separate from an intelligence factor in children as young as 4 and 5. The results indicate that: (1) The ability to implement sequential plans generalizes across tasks rather than being specific to a particular task; and (2) The young children who excel in this ability are not necessarily the same as those with high IQs. In a second factor analysis, the three planning tasks were again included along with the IQ measures, but accuracy measures were substituted in place of the planning measures on these three tasks. When the reanalysis was performed, only one factor emerged, with the planning tasks now loading on the general intelligence factor. This indicates that the original finding, of separate sequential planning and IQ factors, was not an artifact of type of task.

Careful observation of children in classroom settings reveals that competent school functioning requires organization and planning. These skills are clearly important for success in many school tasks (Naglieri & Das, 1987). The more behavior is purposeful and systematic (i.e., governed by a plan), the more effective the children are in the situation. Yet, some children approach tasks with

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haphazard trial-and-error strategies, which typically do not work, or do not work very well. (Bronson, 1981). They approach assignments with little apparent forethought. They may not consider what materials they might need, what kind of workspace would be appropriate, or what series of steps may be required to accomplish the task (Bronson, 1981).

Although planning and organizational skills contribute greatly to success or failure in school (Naglieri & Das, 1987) and later in life, there has been little effort to investigate these abilities in children. The major goals of the research here are to determine whether young children's ability to carry out sequential plans is an ability that generalizes across tasks related to preschool activities, and to determine whether this sequential planning ability is separate from intelligence in young children.

Information-processing theory has provided a general model for studying the mental processes likely to be involved in planning. People are considered to control their behavior through programs, subroutines, and strategies that are stored in long-term memory, are modifiable, and have an effect on a wide range of behaviors (Simon, 1972). These are designated as executive processes. Planning and organization are considered to be key executive processes.

Assessment of Executive Processing Skills

Executive processing skills, such as planning, are not revealed in simple right or wrong answers, but are demonstrated in a person's total approach to a problem or task (Hayes-Roth & Hayes-Roth, 1979; Klahr & Robinson, 1981). Until recently, cognitive assessment procedures have not included ways to measure the strategies involved in the process of thinking. Traditional cognitive testing techniques have focused on content rather than process (Kail & Pellegrino, 1985). They have been concerned with assessing information and knowledge rather than evaluating the ways in which that information is processed.

Recently, methods for assessing information processing have been developed. However, empirical efforts to investigate these processes have been extremely restricted in scope and are very task specific (Neisser, 1976). This presents a problem because executive processes, such as planning and decision making, are very broad based, according to information-processing theorists such as Sternberg (1984), Siegler (1986), and Brown and DeLoache (1978), because they involve generalized rules for problem solving. Neisser (1976) complained that many studies in the field of information processing are "experiments that provide inexperienced subjects with brief opportunities to perform novel and meaningless tasks" (p.8). Neisser argued for a broadening of the information-processing approach to "understanding cognition as it occurs in the ordinary environment and in the context of natural purposeful activity" (p.4).

In contrast to the traditional measurement of strategies in information processing, Bronson (1983) took a more naturalistic approach to studying planning in preschool and early elementary-age children. In order to analyze a wide range of

classroom-planning behaviors, Bronson developed an observation measure, the Executive Skills Profile (Bronson, 1983). This tool uses structured categories to record a child's use of planning strategies in both cognitive mastery tasks and social interactions in the classroom. When used as part of an evaluation battery to predict school achievement, the measure did as well, or better, than other predictors typically used by school systems, such as mother's education and cognitive test scores (Bronson, 1981; Bronson, Pierson, & Tivnan, 1984; Pierson et al., 1983). In this study, similar methods of measuring planning were used, but with tasks specifically designed for more formalized assessment settings (Bronson, Shapiro, Wallace, Desir, & Goodson, 1982; Wallace, Shapiro, Desir, Bronson, & Goodson 1982). The tasks were selected to be similar to activities found in preschool environments. They tapped different parts of the preschool curriculum, including block construction, beginning reading involving story sequencing, and beginning math involving seriation.

Definitions and Measurement of Planning

Planning involves a variety of cognitive and executive skills. In the literature, deciding to plan, knowing when to plan, choosing a goal, formulating a plan, executing, and monitoring plans have been addressed (Friedman, Scholnick, & Cocking, 1987; Scholnick & Friedman, 1987). Furthermore, the motivational aspects of planning in terms of goal-directed behavior (Mook, 1987), and the Piagetian definition of planning as anticipatory schemas (Piaget, 1963) have also been conceptualized. This investigation addresses the way in which sequential plans are *executed* among preschoolers. The component of planning that involves an organized sequence of activities to achieve some goal has been assessed. One reason for using this measure, rather than consciously verbalized plans, is that the use of verbal responses for assessment of planning in young children is likely to underestimate their ability.

In the tasks used in this study, the children were presented with a predetermined goal and asked to reach that goal. The sequences the children used to achieve that goal were then assessed in terms of their planfulness. In some tasks, the planful sequences were prescribed by the task; in other tasks, they were more open-ended, with many planful sequences possible.

The investigation of children's ability to execute plans has generally been studied using tasks which require a sequence of actions. This is because multi-step, sequenced tasks are needed in order to assess the execution of plans in contrast to the one-step tasks frequently used to assess IQ. Search tasks (Gauvain & Rogoff, 1985, Wellman, Fabricius, & Sophian, 1985), mazes (Gardner & Rogoff, 1985), seriation tasks (Welsh, Groisser, & Pennington, 1988), chore-scheduling tasks (Pea & Hawkins, 1987), blockbuilding tasks (Goodson, 1982), and the Tower of Hanoi (Klar & Robinson, 1981; Welsh et al., 1988) are some of the types of problems that have been employed to assess planning in children.

Using a search task, Wellman et al. (1985) found some evidence of planning

behavior in 1- to 3-year-olds. Clear evidence of behavior involving deliberate planning ahead was not found until 4 or 5 years of age. The study here focuses on the older preschool-age group.

There are a number of ways of measuring children's ability to implement plans. Systematic responding has been used as a key measure. For example, in the Tower of Hanoi task, planning can be assessed in terms of the logical order of placement of objects onto pegs versus trial-and-error patterns. It is also possible to assess planning through efficiency of responding. This is frequently the type of measure used in search tasks when the shortest route to a point is assessed.

In this study, several types of tasks involving sequences were employed in order to tap planning in different contexts. One task was a seriation task that measured efficiency. A story sequence task was also included that measured both efficiency in placement of cards depicting the story sequence, and verbal descriptions that were assessed for systematic and logical sequences. Finally, a more complex task was included in the test battery, a block-building task. This task was more open-ended, with a large number of potential strategies for organizing the materials and using efficient and systematic placement of the blocks. Thus, with increasing task complexity in the block-building task, we were able to assess more components of planning execution.

Generality of the Planning Construct in Children. The research here was designed to determine whether there is a general characteristic involving the ability to implement sequential plans in children as young as 4 and 5. In other words, does their ability to execute plans in one sequential task relate to this type of ability in other sequential tasks, with some children generally excelling and other children performing less well?

Although most of the research on planning has been task specific, there is some evidence for the generality of a planning construct in older children. Das (1984a) proposed a general model of planning, and in a study (Das, 1984b) using multiple tasks with third-grade children, eighth-grade children, and retarded adults, a planning factor was identified. In a related study (Das, 1984b), community college students were identified as being good or poor planners, and those previously identified as good planners were able to break codes more quickly and used more appropriate "rational strategies." Those previously identified as poor planners engaged in "superstitious behavior" and did not appear to "appreciate the value of reasoning" in the situation. Kirby and Ashman (1981) also found evidence of general planning factors with fifth-grade children.

Furthermore, Naglieri and Das (1987) found developmental increases in planning ability across three grade levels (2nd, 6th, and 10th), and significant correlations with achievement in reading and math. These findings provide support for the validity of planning ability as a useful and meaningful construct of children's thinking.

Planning as a Separate Factor from Intelligence

In addition to identifying the ability to implement sequential plans as a general construct in young children, this study was designed to determine whether this type of planning ability is separate from general intelligence in this age group. The commonsense view would hold that a separate planning factor should exist, because many individuals are extremely organized without being brilliant, and others are quite bright and yet totally disorganized. Kreitler and Kreitler (1987) showed that when execution of plans were measured in elementary-school-age children (15–11 years) in terms of sequencing (number of steps to solution), there was little correlation with IQ. Using a factor analytic approach, Das and Dash (1983) also showed that among school-age children, sequenced planning tasks formed a separate factor from general intelligence tasks. The question of interest in this study is whether a similar effect can be found among younger children using tasks appropriate to their age level and related to classroom activities.

One limitation to the factor analysis approach is that separate intelligence and planning factors may emerge simply as an artifact of the types of tasks used. In other words, the planning factor may emerge as separate because the *tasks* assessing planning ability are qualitatively different (e.g., multistep problems) from ones used to measure IQ (e.g., one-step problems), not because they are tapping the *process* called sequential planning. One way to investigate this alternative possibility is to score the planning tasks in two different ways. In this study the tasks were first scored to assess sequential planning ability (e.g., organization, systematic responding, and efficiency) and then were rescored to measure accuracy (i.e., number of correct responses on the tasks). It was hypothesized that accuracy measures on these tasks should load on general cognitive ability and, therefore, only one factor should emerge when outcome, not sequential planning ability, is scored. However, when the same tasks are scored for sequential planning ability, two separate factors should be found, with the sequential planning tasks loading on the planning factor, and the general cognitive tests loading on the IQ factor.

METHODS

Subjects

A total of 72 children (33 boys and 39 girls) enrolled in four private preschool programs participated in the study. The schools were located in suburban areas of Boston, MA. The subjects ranged in age from 4 years, 0 months to 5 years, 8 months ($M = 4.88$, $SD = 2.59$, $M = 4.97$, $SD = 5.60$; $M = 5.15$, $SD = 5.26$, $M = 4.95$, $SD = 3.23$ for the four classrooms), and had attended their preschool program for at least 6 months. None of the children had any known handicapping conditions. Each subject was tested individually at the end of the school year in

two separate half-hour sessions, using three trained testers. The tasks were presented in counterbalanced order across subjects. (Two separate testers assessed half the planning and IQ measures for each child.)

Measures to Assess Sequential Planning and Intelligence

Sequential Planning Tasks. The *Animal Stalls task* (Wallace et al., 1982) was designed to be similar to construction and block play activities in early education settings. It involves replicating a block construction from a model. The model is a block structure consisting of "stalls" with miniature animals and hay bales, such as might be found on a farm or in a barn. The child must gather the relevant blocks from a larger array on a small shelf, arrange an appropriate work space, and copy the model exactly. The children are instructed that the construction should look "just like" the original.

The planning construct used in this study for Animal Stalls had three components: organization, systematic approach, and efficiency. While the children performed the task, raters recorded behaviors which fit these three components and also recorded the children's sequence of block placements. Both prior to and during this study, raters reached 90% accuracy in recording children's behaviors and scoring each category.

Categories in the *organization component* demonstrated planfulness by providing evidence of an organized approach to a task. Observers recorded instances of gathering relevant materials, choosing an appropriate work site, and grouping materials in a way that facilitated accomplishment of the task. Each instance was scored as 1 point and then totaled. Children who scored low in this area had difficulty foreseeing the requirements of the tasks. They did not have the right materials at hand, or if the materials were there, they were not arranged in a way to facilitate task accomplishment. The child might have been sitting on needed blocks or have placed them out of sight, or have jumbled materials in a pile so that they were difficult to distinguish. Low-scoring children also typically attempted tasks in spaces which were not big enough, were in the way of something else (like their own feet or a table leg), or were facing the wrong direction (see Table 1 for average scores in this sample).

For the *systematic response component* of planning, categories of behaviors were recorded which indicated a systematically correct approach to the task. The child demonstrated planfulness by receiving a point each time he or she followed a clear and organized sequence of steps in placement of blocks. Children who scored low in this area seemed to be using trial and error rather than a planned approach. One placement of a block bore little relation to the next placement, and there was seldom any discernible direction in these children's activities (see Table 1 for average scores in this sample).

The *efficiency component* recorded the child's number of systematically correct responses in block placements after subtracting moves which were correc-

TABLE 1
Planning Scores on the Three Planning Tasks

Measures	<i>M</i>	<i>SD</i>	Maximum Score Obtained	Maximum Score Possible
Animal Stalls Task				
Organization	1.96	1.80	6	no limit
Systematic response	19.17	10.05	31	31
Efficiency	17.30	9.53	31	31
Seriation Task				
Efficiency	8.01	2.38	10	10
Story Sequence Task				
Systematic response	9.00	5.98	18	18
Storytelling	10.40	7.29	28	no limit

tions of past mistakes. This provided an estimate of how closely the child's performance approximated an optimal performance, that is, the minimum number of trials to solution. Some children scored high on the systematic responding measure by making many errors, which were then systematically corrected. The efficiency measure gave credit to the child who avoided this pattern by making the minimum number of moves necessary to complete the task. The subtraction method eliminated those systematic responses which occurred after a mistake, and counted only those which were correct without having made a prior error. Children who scored low in this area often appeared to have a plan, but seemed to lose track of it at various points or to make careless mistakes. The efficiency component was included so that a child who alternated between working systematically for a while and reverting to trial-and-error responding did not obtain as high a score as a child who was able to complete the task in an orderly way throughout, building up the stalls in a logical manner and clearly showing preplanning (see Table 1 for average scores in this sample).

In order to compute a total sequential planning score for the Animal Stalls task, the scores for organization, systematic approach, and efficiency were combined for a total score (standardized to a Z score). (Distributions for these scores were approximately normal, and Cronbach's Alpha for the total Z score was .89.) In contrast to this sequential planning score, the accuracy measure consisted of the total number of blocks in the correct position at the completion of the task (see Table 1 for average scores in this sample).

The *Seriation task*, designed for this study, required the child to seriate 10 small cylinders with a difference in height of approximately 2.5 cm between

pieces; the materials were similar to toys and manipulatives found in preschool classrooms. To be certain the children understood the task, they were shown the cylinders in a seriated "stair" form, then the pieces were mixed and they were asked to make a stair just like the one they had seen. The planning construct for this task consisted of an efficiency measure of the number of cylinders placed in a sequenced order minus the number of attempts. Thus, trial-and-error responses were not counted (M score = 8.01, SD = 2.38, min = 0, max = 10 for this sample). In contrast to this planning score, the accuracy measure consisted of the number of cylinders placed in the correct sequenced order at the completion of the task (see Table 1 for average scores in this sample).

The *Story Sequence task*, designed for this study, was based on the picture sequence task in the WISC, and is related to cognitive abilities involved when looking at picture books. The children were given six sets involving three pictures, each in random order, and asked to put the pictures in the right order to tell a story, then to tell the story. Both the picture sequence and the verbal account were scored for evidence of planning. Subjects received 1 point for each set for showing evidence of planning if they got the first picture in the correct order on the first attempt. In order to receive additional credit for each set they had to place the next two pictures in correct order (2 points) on the first attempt. They did not get credit for correct placement following trial and error. Again, this allowed for assessment of how systematically the children solved the task (see Table 1 for average scores in this sample). On the storytelling component, the children's responses were audiotaped. Every logically sequenced statement was scored as correct. The statements had to follow logically in relation to the pictures. Subjects who made statements that were lacking in logical sequence or had little relation to the pictures, or both, scored low on this section (see Table 1 for average scores in this sample).

Both the picture-ordering and storytelling scores were added together to form a composite score (standardized to a Z score). (Distributions were approximately normal and Cronbach's Alpha for the composite score was .78.) In contrast to this planning score, the accuracy measure consisted of the number of picture sets out of six having pictures in the correct order at the completion of the task (see Table 1 for average scores in this sample).

The Animal Stalls, Seriation, and Story Sequence tasks were selected to assess sequential planning because they were multistep tasks which could be analyzed to determine whether forethought and planning were used by the children. Many standard assessment tasks are not appropriate for this type of analysis because they are one-step tasks in which the subject either immediately knows or does not know the answer (e.g., comprehension and vocabulary questions on IQ tests). The tasks used to assess planning must allow for evaluation of strategies.

General Intelligence Measures. Three different types of tests were used in order to obtain a measure of general cognitive functioning. Diverse cognitive

tasks were chosen to assess a wide range of cognitive abilities. The Peabody Picture Vocabulary test, which assesses receptive vocabulary, was chosen as measure of verbal agility. The Kaufman Assessment Battery for Children was chosen as the general measure of IQ. It correlates highly with other IQ tests and was designed to measure both simultaneous and successive processing. (Kaufman & McLean, 1987) A Piagetian Number Conservation test was included to measure cognitive development in terms of preoperational and concrete operational thinking.

The *Peabody Picture Vocabulary Test-Revised* (PPVT-R) consists of a series of matching tasks in which the subject has to point to one of four objects on a page in response to a verbal stimulus. This test is a measure of receptive vocabulary and provides an index of English vocabulary achievement. Salvia and Ysseldyke (1981) reported reliabilities of .71 to .89, and indicated that the PPVT-R correlated with standard school performance (for this sample, M score = 113.43, SD = 19.10, min = 68, max = 151).

The *Kaufman Assessment Battery for Children* (K-ABC; Kaufman & Kaufman, 1983; Kaufman & McLean, 1987) defines intelligence as the ability to solve problems using simultaneous and sequential mental processes. The simultaneous processing scale contains task that are spatial or analogic, and the child must simultaneously integrate and synthesize the information in order to solve each problem. For example, in the fact recognition subtest, the child has to pick a face from a group picture; in the triangles subtest, the subject has to assemble several identical triangles into an abstract pattern to match a model. The sequential processing scale presents tasks that the child must solve by arranging the stimuli in sequential order. For example, in the number recall subtest the child must repeat a series of digits, in the word order subtest the child has to touch a series of silhouettes of common objects in the same sequence as the examiner says the names of the objects. Reliability coefficients using the split-half method for the subtests ranged from .86 to .93, and the K-ABC has been shown to correlate highly with other IQ tests and with standard school performance (for this sample, M score = 108.09, SD = 12.95, min = 77, max = 138).

The third measure of general ability was a Piagetian task assessing concrete operational thinking. The *Number Conservation task* was selected. The test was administered using a procedure similar to the one by Piaget described in Inhelder, Sinclair, and Bovet (1974), with the exception that five rather than seven checkers were used in each row (Gelman, 1969). Using this procedure, the children were shown two rows of five checkers, one under the other, with one row red and one row black. When the children agreed that there were the same number of checkers in both rows, the experimenter modified the layout by spacing out the checkers in one of the rows to form a longer row and asked, "Are there as many, the same number of black ones as red ones or aren't there? Or are there more? How do you know?" Then the red checkers were put in a circle, and the black checkers were put in a row, and the question was repeated. A score of 3 was

given when correct responses and explanations were obtained on the two trials, a score of 2 was given when a correct response and explanation was given on only one trial, and a score of 1 was given when subjects responded incorrectly on both trials, showing evidence of preoperational thinking (for this sample, M score = 1.75, SD = .857, min = 1, max = 3, with 52% functioning at the preoperational level, 21% at transition, and 27% at the concrete operations level on this task).

RESULTS

Factor Analyses

A factor analyses was performed on the three sequential planning tasks (Animal Stalls, Seriation, and Story Sequence) using the planning measures, and on the three general intelligence measures (the K-ABC global score, the PPVT-R, and the Piagetian Number Conservation task). The criteria for extracting factors consisted of the Kaiser-Guttman rule of extracting factors with eigenvalues greater than 1. The number of factors extracted was 2.

Varimax rotation was used following extraction of two factors. Results are shown in Table 2. In the factor analysis, the first factor was a planning factor and accounted for 35.5% of the variance. The factors with high loadings included the composite score for the Animal Stalls planning measure, the composite score for the Story Sequence planning measure, and the score for the Seriation planning measure. All other measures had loadings of less than .27.

The second factor was a general intelligence factor accounting for 21.8% of the variance. Results are shown in Table 2 as well. The factors with high loadings included the K-ABC the PPVT-R, and the Piagetian Number Conservation task. All other measures had loadings of less than .33.

The factor analysis was repeated, this time substituting outcome measures in terms of accuracy of performance for the planning tasks in place of planning measures.¹ The results are shown in Table 3. In this reanalysis, including all six tasks, only one factor was obtained, and it accounted for 38.8% of the variance.

DISCUSSION

Planning as a General Ability Separate from IQ

These findings provide support for the view that children as young as 4 can show evidence of an ability to implement sequential plans. Furthermore, this ability is

¹The planning measure correlated .69 with the outcome measure on the Animal Stalls task, .86 on the seriation task, and .82 on the picture sequence task. It would be expected that planning and outcome measures would correlate highly, because good planning should lead to good performance on a task. Despite these strong relationships, there was enough difference in the pattern of correlations that different factor structures were found for the outcome and planning measures when they were factor analyzed in relation to general intelligence. This should not be taken to mean that the two types of measures were unrelated. In fact, they were clearly highly correlated as previously indicated.

TABLE 2
Principal Components Factor Analysis with Varimax Rotation
Using Measures of Planning on the Planning Tasks

	Factor 1	Factor 2
Planning Tasks		
Animal stalls composite planning score	.78	-.17
Seriation planning score	.74	.19
Storytelling composite planning score	.68	.33
Intelligence Tasks		
K-ABC score	.26	.72
Piaget number conservation score	.17	.65
PPVT-R score	-.15	.76

a general one, which extends across a number of school-related tasks. The planning abilities that were assessed consisted of organization (block construction), systematic responding (block construction, seriation, and storytelling), and efficiency (block construction and seriation). Previous research on children of this age has focused on planning in a task-specific context. These results indicate that sequential planning may be a more stable, global characteristic, rather than a narrow phenomenon limited to specific tasks.

These results are of interest because they show that, among children this young, there are already differences in the ability to use organizational and systematic strategies when approaching tasks. Furthermore, those children who excel in this ability are not necessarily those with high IQs, since the findings indicate that ability to implement sequential plans in 4- and 5-year-olds is separate from general intelligence (as measured by a composite of cognitive function-

TABLE 3
Principal Components Factor Analysis with Varimax Rotation
Using Outcome Measures on the Planning Tasks

	Factor 1
Planning Tasks	
Animal stalls outcome score	.51
Seriation outcome score	.72
Storytelling outcome score	.61
Intelligence Tasks	
K-ABC score	.75
Piaget number conservation score	.57
PPVT-R score	.54

ing including a measure of general IQ [K-ABC], vocabulary IQ [PPVT-R], and Piagetian developmental level [Number Conservation task]). This finding provides additional confirmation of Das's (1984a) theory that planning is a separate ability from IQ, based on Luria's (1973) conceptualization of the functional organization of the brain. Prior research on older school-age children (Dash & Dash, 1983; Kreidler & Kreidler, 1987) using different measures of planning and IQ also showed evidence that ability to execute a plan, and general intelligence, form separate abilities. (A note of caution: It is possible that with a less restricted range of subjects including lower SES or retarded children in the sample, higher correlations between planning and IQ would emerge. This remains to be examined in future studies.)

Are the Two Factors an Artifact of Task Selection?

A major contribution of this study was to establish that the evidence for a separate sequential planning factor is not an artifact of the types of tasks used to measure IQ and planning. In the second factor analysis, the same three planning tasks (Animal Stalls, Seriation, and Story Sequence) were again included. However, in this analysis, outcome measures (i.e., number of correct responses) were substituted in place of the planning measures (e.g., organization, systematic approach, and efficiency) for these three tasks. When the reanalysis was performed using these nonplanning measures of performance on the planning tasks, only one factor emerged, with the planning tasks now loading on the general intelligence factor along with the K-ABC, the PPVT-R, and the Piagetian Number Conservation task.

This statistical approach also helps to rule out some of the competing explanations for the findings. For example, in addition to planning, verbal comprehension is necessary for successful performance on the picture sequence task, size discrimination is required on the seriation task, and spatial ability on the Animal Stalls task. Because all of these abilities would affect outcome as well as planning, they cannot be used to provide an alternative interpretation for the separate factor loadings obtained in the initial factor analysis.

The dual method of scoring applied in this study may be a useful tool for other types of research on executive processing skills. Such an approach provides a way of separating the assessment of strategies employed on a task from evaluation of the degree of success or failure achieved when solving the task.

Educational Implications of the Planning Construct

The research on sequential planning presented here suggests that teachers should focus their attention on children's planning skills and consider them as separate from general intellectual ability. Even children this young show a wide range of planning abilities. It remains to be determined whether it is possible to intervene directly to improve planning behavior in children at this age and whether these interventions will have an impact on children's later organization and planning

skills. One advantage of teaching planning strategies during the early years may be to make automatic the processes of asking, "What is my goal for this task, what materials do I need, and what steps will I need to go through to reach my goal?" If young children establish this type of set for using planning strategies when presented with school tasks, it may have long-term benefits, as their developing cognitive abilities allow for more sophisticated and well-developed plans.

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